

## CAIRO, IL, USA

WMO: 724975

Lat: 37.064N

Lon: 89.219W

Elev: 321

StdP: 14.53

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 93809

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 12.4       | 18.1 | 1.7                           | 6.1 | 15.5 | 6.6 | 7.8 | 20.5 | 24.9                  | 44.3 | 22.0 | 42.2 | 5.5                   | 0    | 0.428 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.2                   | 95.4            | 76.0 | 93.1 | 75.8 | 90.8 | 75.5 | 80.2                | 89.7 | 78.9 | 88.2 | 77.8 | 86.8 | 6.7                  | 210  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 77.5                            | 144.9 | 85.3 | 76.5 | 140.0 | 84.1 | 75.2 | 133.7 | 82.5 | 43.9          | 89.4 | 42.6 | 88.0 | 41.4 | 87.2 | 95.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|
| 1%                | 2.5% | 5%   | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |
|                   |      |      | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |
| 19.8              | 17.5 | 15.6 | 4.9                        | 100.0 | 7.0                | 6.8 | -0.1                                               | 104.8 | -4.2       | 108.8 | -8.1       | 112.6 | -13.1      | 117.5 |      |
|                   |      |      | DB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |      |
|                   |      |      | WB                         | 4.5   | 83.2               | 6.9 | 3.6                                                | -0.5  | 85.8       | -4.6  | 87.9       | -8.5  | 89.9       | -13.5 | 92.5 |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 59.3   | 36.4  | 39.4  | 49.8  | 60.0  | 69.6  | 78.4  | 79.7  | 77.8  | 71.2  | 60.0  | 47.9  | 39.9  |
|                                                                                  | DBStd       | 17.64  | 10.84 | 11.01 | 10.28 | 8.79  | 7.64  | 5.07  | 4.82  | 4.93  | 6.38  | 8.90  | 9.23  | 9.62  |
|                                                                                  | HDD50       | 1414   | 439   | 325   | 132   | 19    | 0     | 0     | 0     | 0     | 14    | 147   | 338   |       |
|                                                                                  | HDD65       | 3851   | 888   | 716   | 481   | 195   | 47    | 1     | 0     | 0     | 19    | 209   | 517   | 778   |
|                                                                                  | CDD50       | 4802   | 17    | 30    | 126   | 320   | 607   | 853   | 919   | 862   | 636   | 324   | 82    | 26    |
|                                                                                  | CDD65       | 1763   | 0     | 0     | 10    | 46    | 189   | 404   | 454   | 397   | 205   | 55    | 3     | 0     |
|                                                                                  | CDH74       | 17440  | 0     | 0     | 59    | 371   | 1670  | 4291  | 4785  | 3851  | 1871  | 523   | 19    | 0     |
|                                                                                  | CDH80       | 7258   | 0     | 0     | 5     | 68    | 586   | 1946  | 2143  | 1668  | 698   | 143   | 1     | 0     |
| Wind                                                                             | WSAvg       | 6.0    | 7.7   | 7.6   | 8.1   | 7.8   | 6.2   | 5.2   | 4.2   | 3.6   | 3.8   | 5.3   | 6.2   | 6.7   |
| Precipitation                                                                    | PrecAvg     | 47.9   | 3.0   | 3.6   | 4.7   | 4.6   | 5.2   | 3.7   | 3.8   | 4.1   | 3.5   | 3.0   | 4.1   | 4.4   |
|                                                                                  | PrecMax     | 60.5   | 9.9   | 11.2  | 12.7  | 9.4   | 10.2  | 8.9   | 9.2   | 11.6  | 9.4   | 7.9   | 8.1   | 11.3  |
|                                                                                  | PrecMin     | 29.8   | 0.7   | 1.1   | 1.1   | 1.3   | 1.8   | 0.5   | 0.4   | 0.2   | 0.5   | 0.0   | 1.1   | 0.7   |
|                                                                                  | PrecStd     | 8.3    | 2.2   | 2.2   | 2.7   | 2.3   | 2.3   | 2.1   | 2.1   | 2.4   | 2.2   | 2.1   | 2.2   | 2.8   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 66.4  | 71.6  | 80.8  | 85.7  | 92.9  | 99.3  | 99.9  | 98.5  | 93.0  | 88.2  | 76.8  | 68.2  |
|                                                                                  |             | MCWB   | 61.9  | 62.9  | 64.9  | 68.2  | 70.7  | 73.9  | 77.4  | 77.5  | 75.5  | 72.2  | 63.2  | 63.8  |
|                                                                                  | 2%          | DB     | 63.3  | 66.4  | 74.8  | 81.7  | 89.4  | 95.1  | 95.1  | 94.7  | 90.1  | 84.0  | 72.0  | 64.0  |
|                                                                                  |             | MCWB   | 59.0  | 60.5  | 62.9  | 66.0  | 71.2  | 74.6  | 77.3  | 77.0  | 75.2  | 69.9  | 60.9  | 59.8  |
|                                                                                  | 5%          | DB     | 58.0  | 61.6  | 70.5  | 78.6  | 86.1  | 92.7  | 92.5  | 91.2  | 87.6  | 80.6  | 67.9  | 60.6  |
|                                                                                  |             | MCWB   | 53.0  | 55.3  | 60.8  | 64.4  | 69.2  | 73.7  | 77.3  | 77.1  | 74.2  | 66.6  | 60.2  | 55.6  |
|                                                                                  | 10%         | DB     | 54.0  | 57.0  | 66.0  | 74.9  | 83.1  | 90.1  | 90.1  | 89.2  | 84.2  | 75.4  | 63.5  | 55.2  |
|                                                                                  |             | MCWB   | 49.5  | 51.6  | 57.7  | 63.0  | 68.8  | 73.4  | 76.9  | 76.4  | 72.2  | 64.7  | 55.9  | 51.3  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 62.7  | 65.1  | 67.5  | 71.8  | 75.4  | 79.9  | 82.1  | 82.1  | 78.6  | 74.9  | 68.0  | 64.2  |
|                                                                                  |             | MCDB   | 65.3  | 70.0  | 75.4  | 81.0  | 85.6  | 89.0  | 91.3  | 93.1  | 88.3  | 84.2  | 72.2  | 67.0  |
|                                                                                  | 2%          | WB     | 59.7  | 61.5  | 64.7  | 69.2  | 73.6  | 78.1  | 80.4  | 80.2  | 77.2  | 72.1  | 64.1  | 61.3  |
|                                                                                  |             | MCDB   | 63.0  | 66.7  | 73.1  | 78.0  | 83.5  | 87.9  | 89.5  | 90.2  | 87.1  | 80.3  | 68.9  | 64.1  |
|                                                                                  | 5%          | WB     | 54.7  | 57.3  | 62.0  | 67.0  | 72.2  | 77.0  | 79.3  | 78.8  | 75.7  | 69.9  | 60.7  | 56.5  |
|                                                                                  |             | MCDB   | 57.1  | 60.4  | 69.0  | 75.1  | 81.8  | 87.1  | 88.3  | 88.0  | 84.4  | 76.4  | 65.9  | 59.6  |
|                                                                                  | 10%         | WB     | 49.5  | 51.9  | 59.4  | 64.8  | 70.9  | 75.9  | 78.1  | 77.5  | 74.1  | 67.1  | 56.9  | 51.8  |
|                                                                                  |             | MCDB   | 52.3  | 56.4  | 64.6  | 72.4  | 79.7  | 86.1  | 86.8  | 85.9  | 81.5  | 73.7  | 62.5  | 55.1  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 15.7  | 16.4  | 18.3  | 20.3  | 19.7  | 19.6  | 18.2  | 19.5  | 21.9  | 22.5  | 19.5  | 15.4  |
|                                                                                  |             | MCDBR  | 20.7  | 22.2  | 23.0  | 23.8  | 23.3  | 23.0  | 20.5  | 22.3  | 24.6  | 26.2  | 24.6  | 20.4  |
|                                                                                  |             | MCWBR  | 17.4  | 16.9  | 14.5  | 12.0  | 8.7   | 7.1   | 7.1   | 9.5   | 11.2  | 12.6  | 16.0  | 16.9  |
|                                                                                  | 5% WB       | MCDBR  | 18.3  | 19.9  | 20.2  | 19.1  | 18.9  | 18.2  | 17.7  | 19.6  | 21.5  | 21.3  | 19.5  | 17.8  |
|                                                                                  | MCWBR       | 17.4   | 17.1  | 14.2  | 11.6  | 8.3   | 7.4   | 7.7   | 9.3   | 10.7  | 11.3  | 16.4  | 16.7  |       |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.319 | 0.329 | 0.364 | 0.400 | 0.431 | 0.442 | 0.482 | 0.477 | 0.412 | 0.358 | 0.323 | 0.312 |
|                                                                                  | taud        |        | 2.461 | 2.415 | 2.377 | 2.274 | 2.246 | 2.226 | 2.140 | 2.145 | 2.318 | 2.452 | 2.495 | 2.486 |
|                                                                                  | Ebn at Noon |        | 276   | 287   | 285   | 279   | 271   | 267   | 256   | 254   | 266   | 273   | 272   | 271   |
|                                                                                  | Edh at Noon |        | 27    | 31    | 36    | 42    | 44    | 45    | 48    | 47    | 37    | 30    | 25    | 24    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 695   | 885   | 1224  | 1605  | 1810  | 2041  | 1971  | 1805  | 1512  | 1144  | 782   | 589   |
|                                                                                  | RadStd      |        | 67    | 113   | 107   | 104   | 151   | 128   | 117   | 96    | 91    | 125   | 83    | 52    |

## Historical Trends

|      | DBAvg                   | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) | Station Only            | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (28 neighbors) | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page